YASH RAWAL

yashrawal987@gmail.com | +91-9399159685 | linkedin.com/in/YashRL | github.com/rawal-yash | redriftai.com | Surat, Gujarat, India

PROFESSIONAL SUMMARY

Software Engineering Specialist (Generative AI) specialized in AI Platform Engineering, agentic workflows, and telecom enterprise integrations. Expert in building low-latency multi-agent execution graphs, custom zero-vector-DB tool routing pipelines, and robust RAG infrastructure. Experienced in working with large enterprise platforms (including tier-1 telecom providers), fine-tuning domain-specific embedding models, and implementing open-source contributions to Spring AI.

TECHNICAL SKILLS

Agentic Systems & Orchestration: LangGraph, LangChain, Multi-Agent Systems, ReAct Execution Loops, Custom Tool Routing, Model Context Protocol (MCP), n8n.
AI Modeling & Fine-Tuning: Supervised Fine-Tuning (SFT), LoRA, PEFT, E5-large-v2, Triplet Loss, Embeddings, PyTorch, Hugging Face Transformers.
AI Evaluation & MLOps/LLMOps: LLM-as-Judge, Golden Test Scenarios, Regression Detection, Batch Scenario Testing, Prompt Lab, Docker, Kubernetes.
Backend & Infrastructure: Python, Java, Spring Boot, Spring AI, FastAPI, PostgreSQL/pgvector, REST/HTTP APIs, Server-Sent Events (SSE).
Security & Guardrails: Phantom Token Security Pattern, PII Sanitization, Adversarial Red Teaming, Observer Guardrails.

WORK EXPERIENCE

TalenTeam Ltd., UK (via Bigscal Technologies)

AI/ML Engineer (Platform Architect focus) — Oct 2024 – Present

Surat, Gujarat, India (Remote)

- Architected and scaled an enterprise AI capability platform on SAP BTP serving over one million active users across more than 120 organizations, including a tier-1 telecom provider.
- Optimized a hierarchical multi-agent system by implementing a flat capability-sharing contract model, which consolidated sub-agent toolsets into the main execution loop, reducing production latency by 3x and token consumption by 70%.
- Developed a zero-vector-DB tool routing layer using BM25F, TF-IDF, and Latent Semantic Analysis fallback, scaling routing capacity to over 1,000 enterprise tools with minimal latency.
- Built Prompt Lab, an offline evaluation control plane using FastAPI and an LLM-as-judge scoring pipeline to run batch evaluations on 500+ golden scenarios, decreasing QA verification cycles tenfold.
- Implemented a hierarchical RAG pipeline for over 10,000 technical assets, utilizing Grobid for layout extraction and a custom PostgreSQL paragraph signature hashing algorithm to prevent redundant storage on re-uploads, achieving a 90% Recall@10 rate.
- Designed the Phantom Token security pattern to pass authenticated user credentials securely between custom agents and Model Context Protocol servers without exposing raw JWT payloads.
- Engineered a Server-Sent Events streaming protocol to deliver token streams, reasoning trajectories, and dynamic UI widget components to the frontend in real time.

Healthray (Bigscal Technologies)

AI/ML Engineer Intern — Apr 2024 – Sep 2024

Surat, Gujarat, India

- Built multilingual clinical transcription and discharge summarization workflows using Whisper and custom medical named entity recognition rules.
- Quantized Llama 3 8B to 4-bit locally using llama.cpp and PyTorch/Hugging Face transformers to optimize local GPU inference constraints.
- Fine-tuned text-to-speech models using the LJSpeech format and custom noise-reduction filters, boosting processing throughput by 60%.

OPEN SOURCE & MODEL TRAINING PROJECTS

Spring AI (spring-projects/spring-ai)

Active Contributor — Mar 2026 – Present

- Fixed metadata parity in the OpenAiChatModel class (PR #5711) by restoring the missing reasoningContent field in the non-streaming internalCall method, enabling compatibility for DeepSeek-R1 and Qwen reasoning models.
- Contributed Amazon Bedrock Converse API prompt-cache time-to-live configuration parameters.

E5-large-v2 Job & Skills Embedding Fine-Tuning

Lead Developer — Jul 2026

- Fine-tuned a 335M-parameter E5-large-v2 model on Lightcast labor-market data using PyTorch and triplet loss (anchor + positive + hard negative), raising semantic retrieval relevance by 18% over OpenAI text-embedding-3-small.

Qwen2.5-7B Supervised Domain Fine-Tuning

Lead Developer — Jun 2026

- Trained a specialized 7B-parameter Qwen2.5 model for structured Job Architecture generation, fine-tuning on 10,000+ synthetic instruction-response pairs via LoRA (PyTorch/Transformers) to enforce schema compliance.

EDUCATION

Parul University — 2022 – 2024

Master of Computer Applications (MCA) – Specialization: AI & Machine Learning

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900) — 2024
Microsoft Certified: Azure Fundamentals (AZ-900) — 2024